YWD University Design Project and Lab Exam

Overview

Your team will design, defend, build and test a redundant campus network for a YWD University, using the equipment in your rack.

Week	Tasks	Due
1	Work from the scenario and requirements below to design an effective solution.	Preliminary design, due before the Week 2 session
2	Defend your design to the instructors and start building.	Defense (15-20 minutes)
3	Finish the build. Instructors test it and try to break it. The business will also send a change request.	A working network, and the change request check-off

Scenario

YWD University is opening an engineering campus and wants a network that is fast, safe, able to grow and hard to take down. Your team is the network design group. The campus has three buildings/locations:

- **Building A:** teaching labs and student work areas.
- **Building B:** faculty and administrative offices.
- **Server Room:** hosts the campus's applications, including a student-portal web app that must be reachable by hostname.

All traffic leaving the campus goes through the two edge routers. We don’t have a real internet connection in the lab, so each edge router has a loopback address that stands in for "the Internet". Reaching either one will count as reaching “the Internet”.

YWD University will add more users and labs over the next two years. A design that has to be rebuilt to grow will lose points.

Business requirements

YWD University has four requirements, each with targets you must meet and be able to demonstrate.

Requirement	Targets
Performance	• The server room has at least 4 Gbps of capacity toward the rest of the campus. • Redundant links should share the load where your design allows it, so no link is blocked and idle. • Reduce or eliminate unnecessary traffic detours..

Requirement	Targets
Security	• Guest/IoT users can reach only the DHCP and DNS services and the Internet • Students and faculty can reach the web app on TCP 80 and 443 and the Internet • Students/faculty cannot reach each other's VLANs.
Scalability	Concerns: • Growth in student + faculty counts • Route summarization (no more than one static route to each building) We expect growth in student and faculty count in our new campus. At what point will we need more hardware?
Reliability/High Availability	• Single point of failure should not cut any user group off from the Server Room or the Internet • Recovery from failures should be fast.

Every choice in your design should trace back to one of these requirements. ACLs are required.

Rules:

- **Routing:** static routes and a first-hop redundancy protocol (HSRP or VRRP) only. No dynamic routing protocols.
- **Backups:** keep a text copy of the configs you put on the device's and save to startup config before you leave each session. Do not worry about resetting to the original startup configuration.
- **Pi's TempFS:** The temporary filesystem on the pi's will be turned off for simplicity. This means if you break something on a pi somehow, it is up to you to resolve it.

User groups and address space

You have 10.0.0.0/16 to work with, and at least half of it must stay unallocated.

Group	Location	Devices today	Bandwidth notes
Students	Building A	120 users	About 5 Mbps average per user.
Faculty and staff	Building B	60 users	About 5 Mbps average per user.
Guest and IoT	Building A (50) and Building B (30)	80 devices	Insignificant
Servers	Server Room	8 servers	The student portal must serve up to 400 concurrent users at about 10 Mbps each,

Group	Location	Devices today	Bandwidth notes
			from inside the campus and from the Internet through the edge routers. That is 4 Gbps toward the Server Room.

Use your Raspberry Pis and network namespaces to stand in for users and servers. Put at least one test host in every VLAN group in every building where that group exists, so you can prove reachability and the access rules.

DHCP and DNS

User devices must get their addresses automatically, and the portal must be reachable by name. We have not covered these yet, so part of your design work is learning how they work and deciding where they live.

DHCP

- At least guests, students and faculty get addresses from DHCP pools on your Cisco devices.
- Each pool must hand out the correct default gateway and a DNS server, and must not hand out reserved addresses.

DNS

- You do not need a full DNS server. An IOS device can answer DNS queries from a static host table.
- The portal is `portal.ywd.com`, pointing at the web app in the Server Room.
- Clients learn the DNS server address from DHCP, and you should prove the name resolves from a test host in every user VLAN.

Student Web Portal

Run the provided portal.py inside a network namespace on a Pi in the Server Room VLAN.

Design Packet Portions

#	Deliverable	Must include
1	Topology diagram	Logical and physical views, with every device assigned to Building A, Building B or the Server Room
2	Addressing plan	VLSM subnets carved from 10.0.0.0/16 for each VLAN, with gateway, FHRP virtual IP and DHCP range
3	Layer 2 plan	VLANs, trunks and allowed VLANs, the link aggregation group, and the STP root and backup root for each VLAN
4	Layer 3 plan	In what network device(s) do the gateways live? FHRP groups with priorities and any tracking, the static routes, and the default route
5	Access policy	A matrix of which groups may reach which, and the ACLs that enforce it, including where each ACL is applied and in which

#	Deliverable	Must include
		direction
6	Services plan	DHCP pools and relay, the DNS entries, and where the web app runs
7	Requirements trace	For each target in the business requirements, the design choice that meets it
8	Verification & Test plan	How you will prove each requirement, including the failures you will simulate and what you expect to happen

Grading Rubric

This is worth 50% of the lab exam grade.

Component	Points	What is assessed
Design package	25	Completeness, correctness, and a clear explanation from each requirement to design choices (tradeoffs!)
Design defense	15	How well the team explains and justifies its choices when questioned
Build and configuration	25	The build matches the design and is functional
Testing and failure handling	20	Requirements shown to work, and how the network behaves when instructors break things
Week 3 Change Request	15	A check-off in Week 3 that the new business requirement works

Design Defense You have 15-20 minutes. Spend about five-ten minutes presenting your design, and we will spend the rest questioning it. We will ask questions to a random team member, who may phone a friend only once. If one person cannot explain a decision, **it counts against the whole team**, so make sure all of you understand all of it!

Build Test

You will run your test plan for us, then we will run our own tests.

Command patterns

A namespace host on a Pi, attached to a VLAN

```
sudo ip link add link eth0 name eth0.<vlan> type vlan id <vlan>
sudo ip link set eth0.<vlan> up
sudo ip netns add <name>
sudo ip link add link eth0.<vlan> name <name>-v type macvlan mode bridge
sudo ip link set <name>-v netns <name>
sudo ip netns exec <name> ip link set <name>-v up
sudo ip netns exec <name> dhclient -v <name>-v
```

IOS DHCP, DNS

```
ip dhcp excluded-address <first> <last>
ip dhcp pool <NAME>
   network <network> <mask>
   default-router <gateway>
   dns-server <dns address>

ip dns server
ip host <Fully Qualified Domain Name > <ip address>
```

ACL

```
ip access-list extended <NAME>
   permit <protocol> <source> <destination> eq <port>
   deny ip any any
interface Vlan<id>
   ip access-group <NAME> in
```

Every ACL ends in an implicit deny. An ACL can block DHCP requests and DNS queries unless you permit them, so think about which traffic your clients need before you apply it.

```
ip dhcp excluded-address 10.0.0.1
ip dhcp pool TEST
   network 10.0.0.0 255.255.255.0
   default-router 10.0.0.1
   dns-server 10.0.0.1

ip dns server
ip host portal.campus.test <web app address>
```